Data Analysis Project – Student Performance using Pandas

InternSpark Internship Task 3

1. Project Objective

The objective of this project is to analyze student marks using Pandas. The program calculates total marks and averages, classifies performance, finds the class average and top performer, and creates a chart.

2. Technologies Used

Python 3, Pandas and Matplotlib. The project was executed using Google Colab on Android.

3. Analysis Performed

• Student marks are loaded into a Pandas DataFrame.

• Total marks and average marks are calculated.

• Each student is assigned a performance category.

• The class average is calculated.

• The student with the highest average is identified.

• The analyzed data is exported to a CSV file.

• A bar chart of student averages is generated.

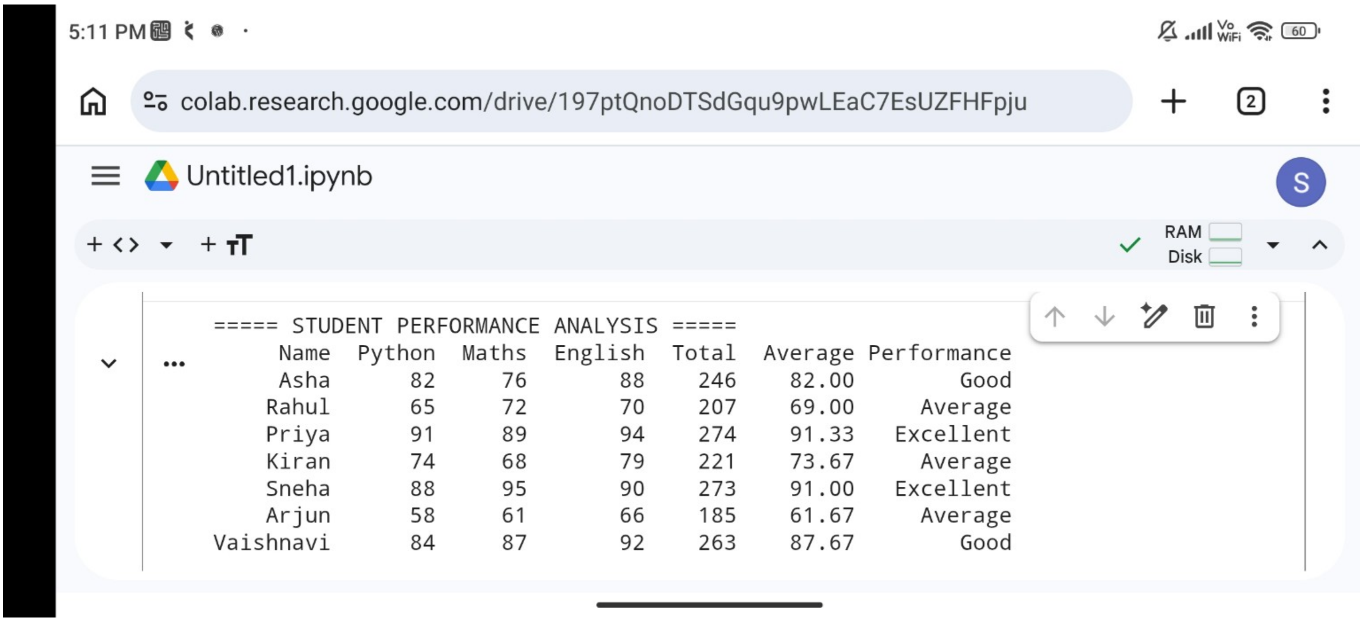
4. Source Code

```
import pandas as pd
import matplotlib.pyplot as plt
from io import StringIO

data = """Name,Python,Maths,English
Asha,82,76,88
Rahul,65,72,70
Priya,91,89,94
Kiran,74,68,79
Sneha,88,95,90
Arjun,58,61,66
Vaishnavi,84,87,92
"""

df = pd.read_csv(StringIO(data))

df["Total"] = df[["Python", "Maths", "English"]].sum(axis=1)
df["Average"] = df[["Python", "Maths", "English"]].mean(axis=1).round(2)

def category(average):
    if average >= 90:
        return "Excellent"
    elif average >= 75:
        return "Good"
    elif average >= 60:
        return "Average"
    return "Needs Improvement"

df["Performance"] = df["Average"].apply(category)

print("===== STUDENT PERFORMANCE ANALYSIS =====")
print(df.to_string(index=False))

class_average = df["Average"].mean()
print("\nClass Average:", round(class_average, 2))

topper = df.loc[df["Average"].idxmax()]
print("Top Performer:", topper["Name"])
print("Topper Average:", topper["Average"])
df.to_csv("student_analysis_result.csv", index=False)
```

```
plt.figure(figsize=(8, 5))
plt.bar(df["Name"], df["Average"])
plt.title("Student Average Marks")
plt.xlabel("Student")
plt.ylabel("Average Marks")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("student_average_chart.png")
plt.show()

print("\nAnalysis completed successfully!")
print("CSV file and chart have been created.")
```

5. Execution Evidence – Analysis Table

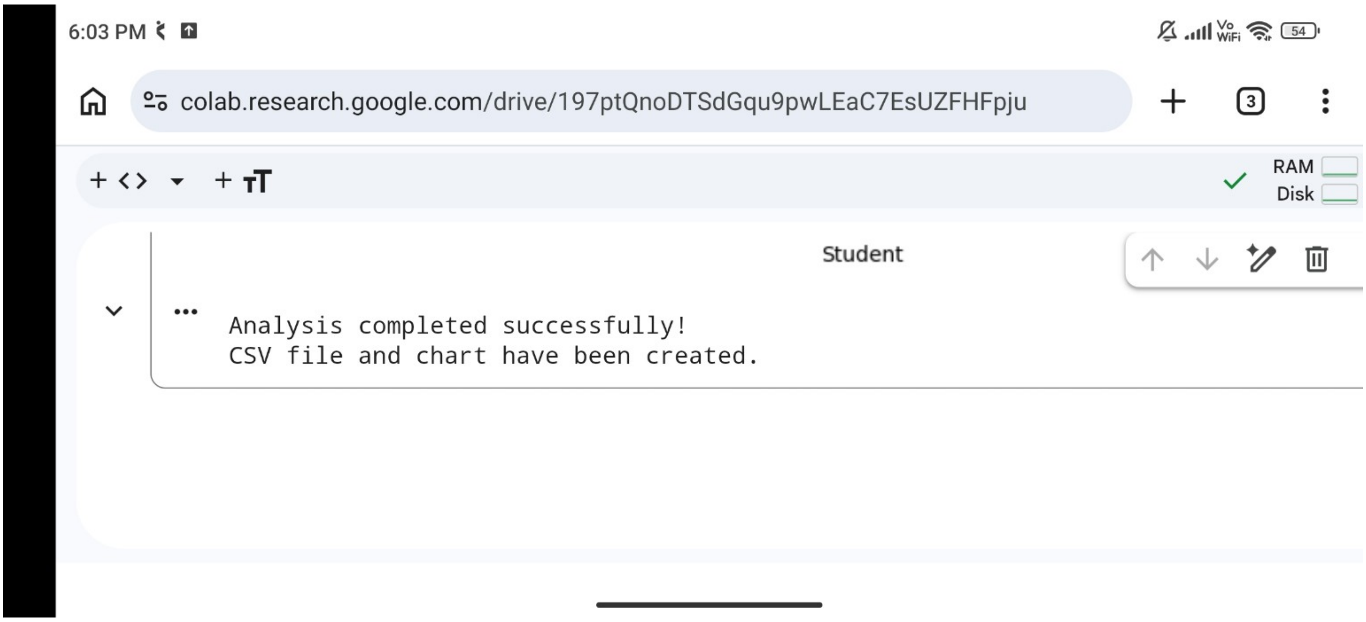
The screenshot below is the actual Google Colab output showing the student performance table.



Name	Python	Maths	English	Total	Average	Performance
Asha	82	76	88	246	82.00	Good
Rahul	65	72	70	207	69.00	Average
Priya	91	89	94	274	91.33	Excellent
Kiran	74	68	79	221	73.67	Average
Sneha	88	95	90	273	91.00	Excellent
Arjun	58	61	66	185	61.67	Average
Vaishnavi	84	87	92	263	87.67	Good

6. Execution Evidence – Summary and Chart

The second actual screenshot shows the class average, top performer, topper average, bar chart and successful completion message.



7. Result

The analysis completed successfully. The class average shown in the execution was 79.48 and Priya was identified as the top performer with an average of 91.33. The CSV result and student average chart were also created.

8. GitHub / Public Project Link

Add the real public GitHub repository link here after uploading the Task 3 code.

GitHub Repository: ____

9. Conclusion

This project demonstrates practical data analysis with Pandas and visualization with Matplotlib. It converts raw student marks into useful performance information and a visual chart.